

APL Logistics Profitability Analysis

Project Report

Abstract

This project presents an interactive business analytics dashboard for analyzing the profitability and operational performance of APL Logistics order data. The application was developed using Python and Streamlit, with Pandas and NumPy for data processing and Plotly for visualization. The dashboard examines sales, profit, customers, products, categories, discounts, markets, countries, regions, shipping modes, and delivery performance through interactive filters and analytical views. The dataset contains 180,519 order records and 40 columns.

1. Introduction

Logistics businesses generate large volumes of transactional data. Converting this data into actionable insights requires systematic analysis of revenue, profitability, customer behavior, products, discounts, markets, and shipping operations. This project provides a centralized interactive dashboard for exploring these dimensions.

2. Problem Statement

Raw logistics transaction data can make it difficult to quickly identify profitable customers, products, categories, markets, and shipping modes. High sales do not always correspond to strong profitability, while discounts can affect profit performance. The project addresses this through interactive profitability and business analysis.

3. Objectives

Analyze total sales, total profit, profit margin, and order volume; identify high-value and low-value customers; compare customer segments; analyze product and category profitability; identify high-revenue markets with relatively weak margins; analyze discount levels against profitability; provide what-if discount scenarios; compare markets, countries, regions, and shipping modes; and provide interactive filters.

4. Dataset

The APL Logistics dataset contains 180,519 records and 40 columns. Important fields include Sales, Order Profit Per Order, Order Item Discount, Order Item Discount Rate, Order Item Profit Ratio, Product Name, Category Name, Market, Order Region, Order Country, Customer Segment, Shipping Mode, and Delivery Status.

5. Technology Stack

Python, Streamlit, Pandas, NumPy, Plotly, Git, and GitHub.

6. Methodology

The workflow loads and prepares the dataset, applies user-selected filters, aggregates sales/profit/order metrics, generates customer/product/category/discount/market/regional/shipping analyses, and presents results through Streamlit tabs, KPI cards, charts, tables, and interactive controls.

7. Dashboard Modules

Executive Overview: Total Sales, Total Profit, Profit Margin, Total Orders, and Average Order Profit. Customer Analysis: customer value, top/bottom customers, Customer Value Index, and segment performance. Product

and Category Analysis: top/bottom products, product margins, category profitability, heatmap, loss-making categories, and high-revenue/weak-margin analysis. Discount Analysis: average discount, total discount amount, profit ratio, discounted orders, discount-rate analysis, discount impact ratio, and what-if analysis. Market and Shipping: market, country, region, sales-vs-profit, weak-profit market, and shipping-mode analysis.

8. Interactive Filters

The dashboard provides filters for Market, Customer Segment, Category, Product, Shipping Mode, Order Region, and Discount Rate. Changing filters dynamically updates dashboard calculations and visualizations.

9. Results and Insights

The dashboard supports monitoring overall financial performance, comparing customer segments, evaluating product and category profitability, examining discount levels, identifying high-revenue markets with weaker margins, comparing geographic performance, and evaluating shipping modes. Under the full dataset view, the dashboard displays approximately \$36.78 million in sales, \$3.97 million in profit, a 10.78% profit margin, and 180,519 orders.

10. Conclusion

The APL Logistics Profitability Analysis dashboard transforms logistics transaction data into an interactive business analytics solution. It combines financial, customer, product, discount, geographic, and shipping analysis in one Streamlit application and demonstrates practical Python-based data analytics, visualization, KPI design, filtering, aggregation, and business-oriented interpretation.

11. Future Scope

Possible enhancements include predictive profitability modeling, customer churn prediction, demand forecasting, automated anomaly detection, advanced discount optimization, role-based access, automated reports, and integration with live logistics data sources.

12. Project Links

GitHub: https://github.com/sayan-3139/APL_Logistics_Profitability_Analysis

Live Dashboard: <https://apllogisticsprofitabilityanalysis-bftqjhwtfyzky6t4kjlwv.streamlit.app/>

References

1. Streamlit documentation: <https://docs.streamlit.io/>
2. Pandas documentation: <https://pandas.pydata.org/docs/>
3. Plotly Python documentation: <https://plotly.com/python/>